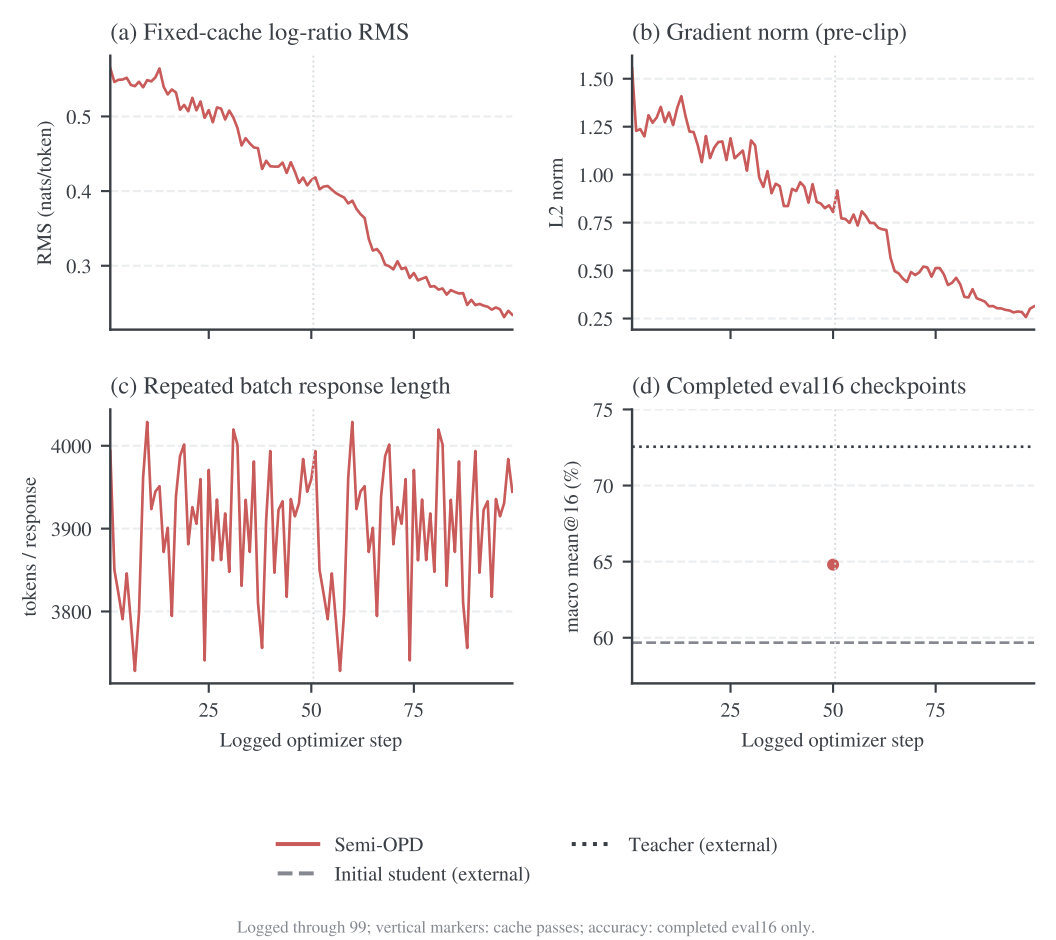
(a) Fixed-cache log-ratio RMS
RMS (nats/token)
0.5
0.4
0.3
(b) Gradient norm (pre-clip)
L2 norm
1.50
1.25
1.00
0.75
0.50
0.25
(c) Repeated batch response length
tokens / response
4000
3900
3800
25
50
75
Logged optimizer step
(d) Completed eval16 checkpoints
macro mean@16 (%)
75
70
65
60
25
50
75
Logged optimizer step
Semi-OPD
Initial student (external)
Teacher (external)
Logged through 99; vertical markers: cache passes; accuracy: completed eval16 only.